# Cart lab: change driving force

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use controlled data to describe how net force affects acceleration.

**SUCCESS CRITERIA - I can...**

1. Change driving force while keeping total mass fixed.
2. Use data or a graph to compare accelerations.
3. Explain how resistance affects the ideal relationship.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

A rover travels straight east at a steady 6 m/s. A second rover turns at the same steady speed. Which can have zero net force? Explain using velocity.

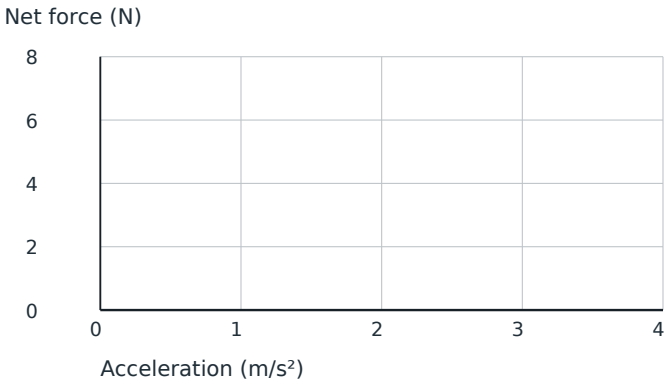
**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $a = F_{net}/M$; $F_{drive} = m_h\, g$.
Keep total moving mass M fixed. The driving weight equals net force only when resistance is negligible.

**Worked reasoning:** A 2 kg cart has net forces 2 and 6 N in two trials. Using a=F/m gives accelerations 1 and 3 m/s². Tripling net force triples acceleration.

**Your turn:** Draw a force-versus-acceleration graph for these trials. Explain the meaning of the slope.




**Marzano check - circle:** ① ② ③

## 03 Read a proportional pattern

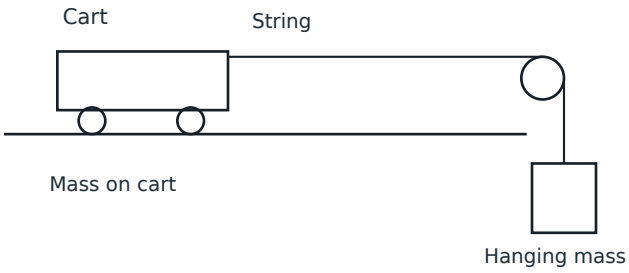
Use the supplied model data for the same 2 kg system. What happens to acceleration when force doubles? Predict a at 8 N.

| Net force (N) | a(m/s²) |
|---|---|
| 2 | 1 |
| 4 | 2 |
| 6 | 3 |

**Marzano check - circle:** ① ② ③

## 04 Keep mass constant

A student increases the hanging mass by adding a mass from the supply box. Use the setup model to identify the two quantities this changes. Describe how to change driving force while keeping total moving mass constant.




**Marzano check - circle:** ① ② ③

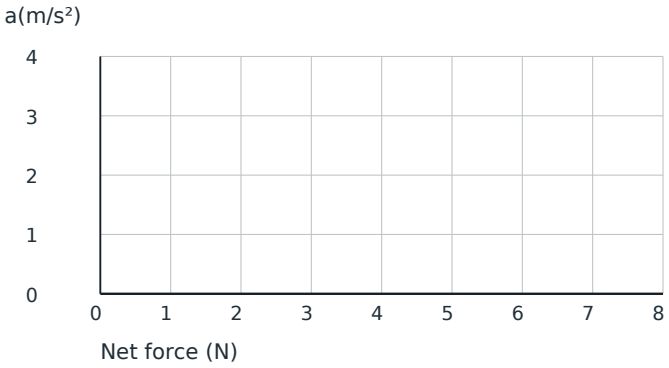
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## 05 Graph the pattern

Plot the model data from 3. Label both axes and draw a trend line. If you collect your own data, put those in a separate table and label them measured.




**Marzano check - circle:** ① ② ③

## 06 Review: connect an earlier idea

With a fixed net force, a 3 kg system accelerates at 10 m/s². Predict acceleration for a 6 kg system. State a control needed for this comparison.
Use the table to keep the controlled and changed quantities distinct.

| Trial | Mass | Net force | Acceleration |
|---|---|---|---|
| A | ___ | ___ | ___ |
| B | ___ | ___ | ___ |

**Marzano check - circle:** ① ② ③

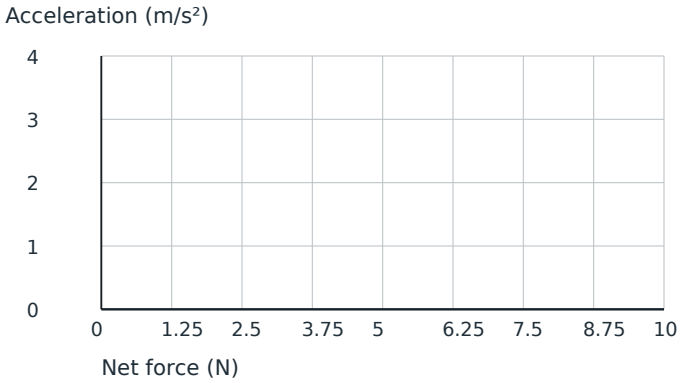
## 07 Exit check

A fixed-mass system accelerates at 2 m/s² under 4 N net force. Predict its acceleration under 6 N net force. Explain with a comparison.

**Marzano check - circle:** ① ② ③

## 08 STRETCH: test the model

Design three net-force settings that give accelerations 1, 2, 3 m/s² for a 3 kg cart. Explain what must stay fixed.




**Marzano check - circle:** ① ② ③